

Changyu (Thomason) Zhao

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EDUCATION

University of Wisconsin-Madison

Sep. 2021 – May 2025

Bachelor of Science in Computer Science (Honors) and Math

Madison, Wisconsin

Cumulative GPA: **3.97/4.0**, **Dean's List** 2021 - 2023

EXPERIENCES

Tencent Xuanwu Lab

Aug. 2023

Security Researcher Intern

Beijing, China

- Engineered a cross-platform (Linux and Windows), **hardware-assisted dynamic control flow analysis** tool in **C&C++** that leverages Intel's Last Branch Record (LBR) technology for enhanced system evaluation.
- Designed and implemented custom **kernel modules** and **device drivers** to enable LBR feature for Linux & Windows.
- Architected and developed a user-mode analysis library that leverages collected LBR to accurately **reconstruct the control flow graph (CFG)** of executed programs. The library incorporated features for parsing, analysis, and visualization.

UW-Madison Cybersecurity Operation Center

Feb. 2023 – Present

Cybersecurity Student Intern Analyst

Madison, Wisconsin

- Managed security incidents, including respond to alerts, investigate security breaches, and take corrective action.
- Use **Elastic** and **Palo Alto Firewall** to monitor and **analyze campus network traffic**, to identify potential threats and vulnerabilities and notify security alert to related users or departments.
- Conducted **Identity and Access Management (IAM)** for access restrictive full-time and student employee accounts.
- Processed **Daily Threat Intel** with team members, fostering a **collaborative environment** to address security concerns and mitigate potential threats in campus.

ASU SEFCOM/CDF Security Research Apprenticeship

May. 2023 – Aug. 2023

Research Assistant

Tempe, Arizona

- Conceptualized, designed, and implemented LibHopper, a **Python-based** security research tool designed to **automate the discovery of exploitation primitives** within arbitrary user libraries.
- Leveraged automated processes to pipeline the evaluation of security implications related to **memory corruption vulnerabilities** in library internal structures, enhancing the capabilities of security researchers and developers.
- Employed a multifaceted methodology involving **customizing debugger scripts** and **symbolic execution** techniques to conduct comprehensive exploitability analysis.
- Identified and cataloged **assets of exploitable primitives** within library internal states, derived from meticulous analysis of memory corruption scenarios and symbolic execution findings on 6 most widely-used C standard libraries.

PROJECTS

Router automated vulnerability discovery | *Reverse Engineering, Firmware, Python*

May 2022 – Aug. 2022

- Exploited unfixed Common Vulnerabilities and Exposures (CVE) in **Tenda AC18 router firmware**.
- Employed **qemu** to rehost arm32 router firmware on an x86-64 machine, enabling systematic debugging.
- Developed **automated** stack overflow and command injection detection IDA plug-in at decompiled code level.
- Achieved Remote Code Execution (RCE) by **exploiting** vulnerabilities identified by the IDA plug-in.
- Discovered multiple unfixed CVEs: CVE-2018-16334, CVE-2018-5767, CVE-2018-5768, CVE-2018-5770, CVE-2020-15916, CVE-2022-28556, CVE-2022-28557.

EXTRACURRICULAR ACTIVITIES

r3kapig CTF Team

Sep. 2021 – Present

Team member

China

- Achieved a notable accomplishment by securing a rank 12/1282 worldwide in the DEFCON 30 CTF Finals (2022), and 7/1828 worldwide in the DEFCON 31 CTF Finals (2023).
- Participated in over 40 48-hour Capture the Flag (CTF) competitions since 2021, with a focus on Binary Exploitation (PWN) and Reverse Engineering challenges.

pwn.college

Dec. 2021 – Jul. 2022

Independent Learning

Tempe, Arizona

- Attained the Yellow Belt and Blue Belt levels by completing **all** challenges in Dojo instance, currently ranked 43/2500+.
- Acquired comprehensive skills in various areas of **software and system security**, including program interaction, shellcoding, privilege escalation, reverse engineering, stack corruption, heap corruption, race conditions, and kernel security.

TECHNICAL SKILLS

Proficient in reverse engineering, have reverse experience in large-scale projects. **Familiar** with control flow tracking, data flow tracking, symbolic execution and the use of binary analysis tools.

Programming Languages: C/C++, Python, Java, Assembly Language, JavaScript, HTML/CSS, $\LaTeX$

Binary Security Tools: IDA/Ghidra, GDB/Windbg, Angr, Qemu, Frida, Joern

Web Security Tools: Burpsuit, Wireshark, Metasploit Framework, Kali Linux

Soft Skills: Time/Project Management, Collaboration, Conflict Resolution, Active Listening